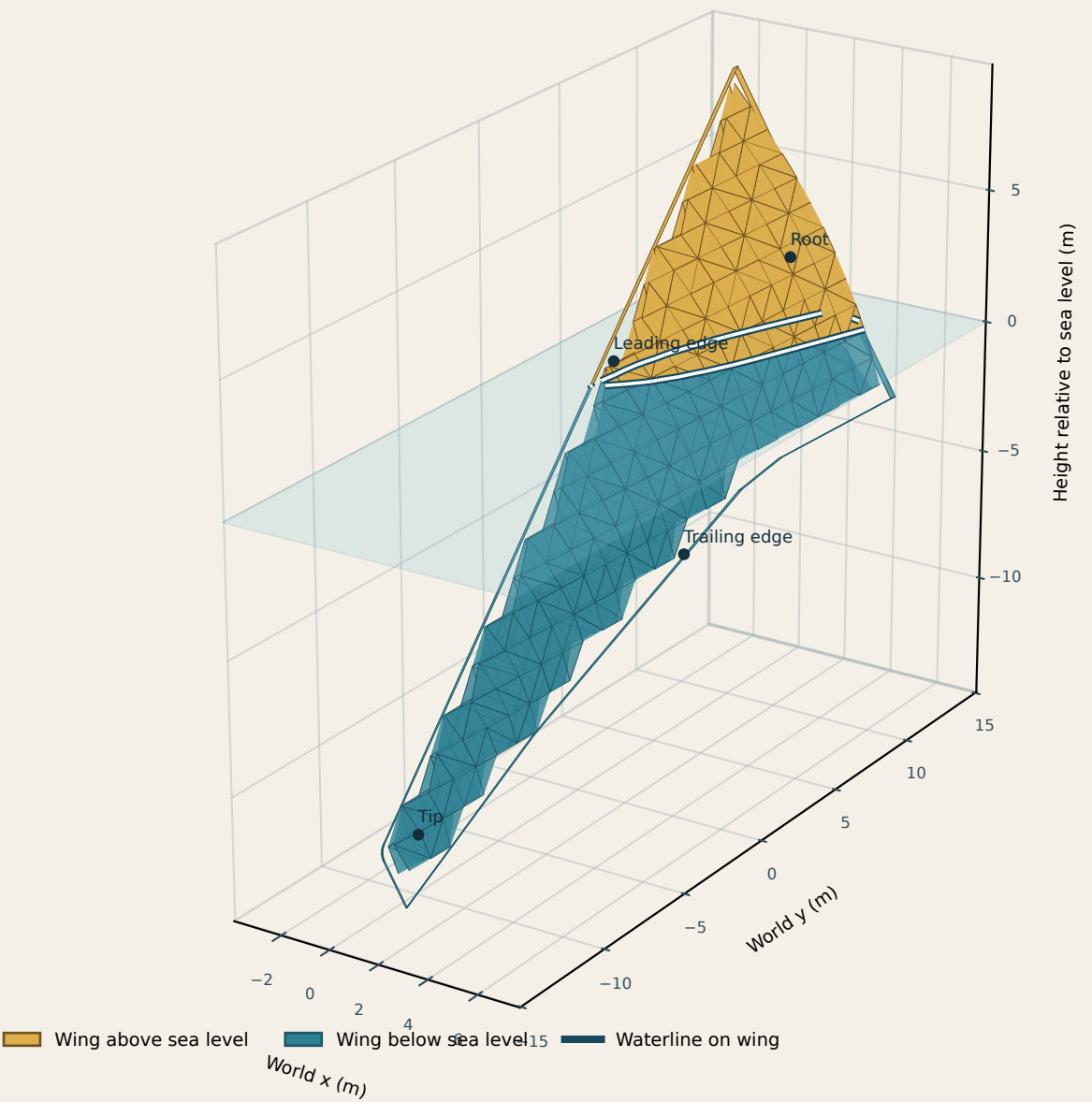


PHR_4_18 — First-Order Complete-Wing Flooding Sensitivity

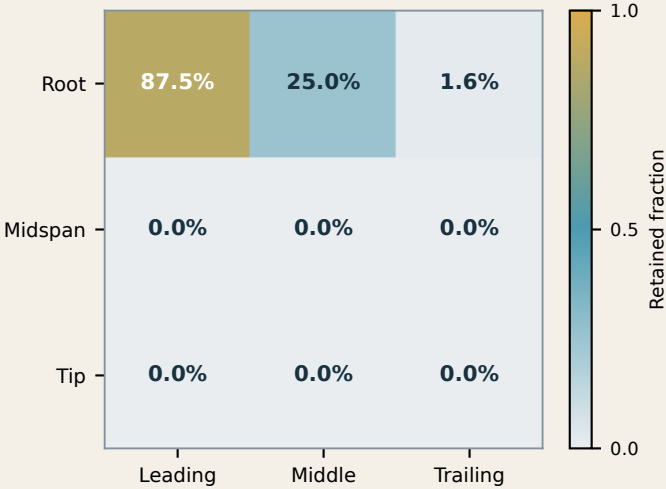
Comparison with the exact PCA-selected random pose; the level sea surface and modeled waterline are explicit.

Closest Target-Like Hydrostatic Equilibrium

Roll +19°, pitch +59°, tilt 60.9°

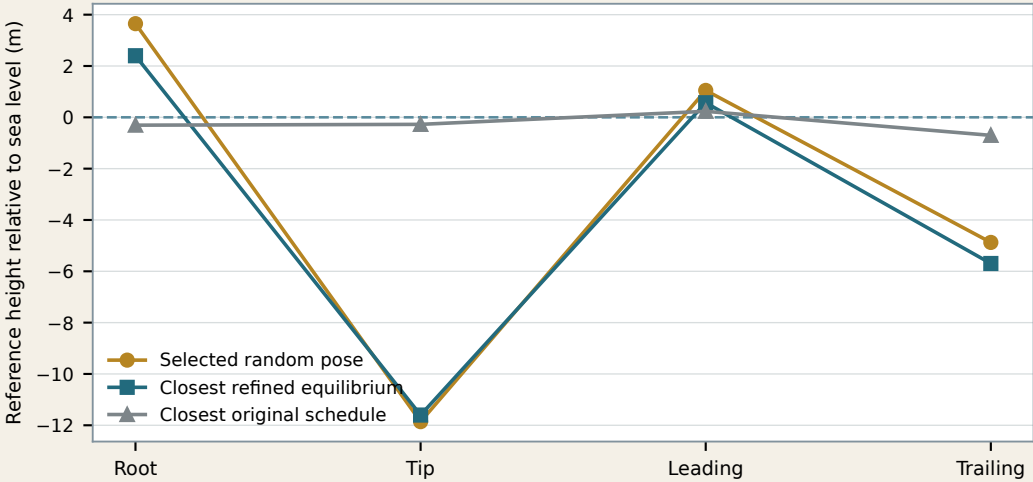


Modeled Retained-Air Fraction by Sensitivity Cell



10/24 refined states met the descriptive attitude and strict size criteria; all had the engine absent and all six midspan/tip cells fully flooded.
Selected state: height RMSE 0.796 m; PCA loss 0.3421; modeled retained air 49.3 m³.

Waterline-Relative Reference Heights



Post-hoc conditional sensitivity, not a flooding probability or global continuous optimum. The 3 × 3 cells and modeled air volume are coarse equivalent-buoyancy assumptions, not documented Boeing tank bays or tank volume.